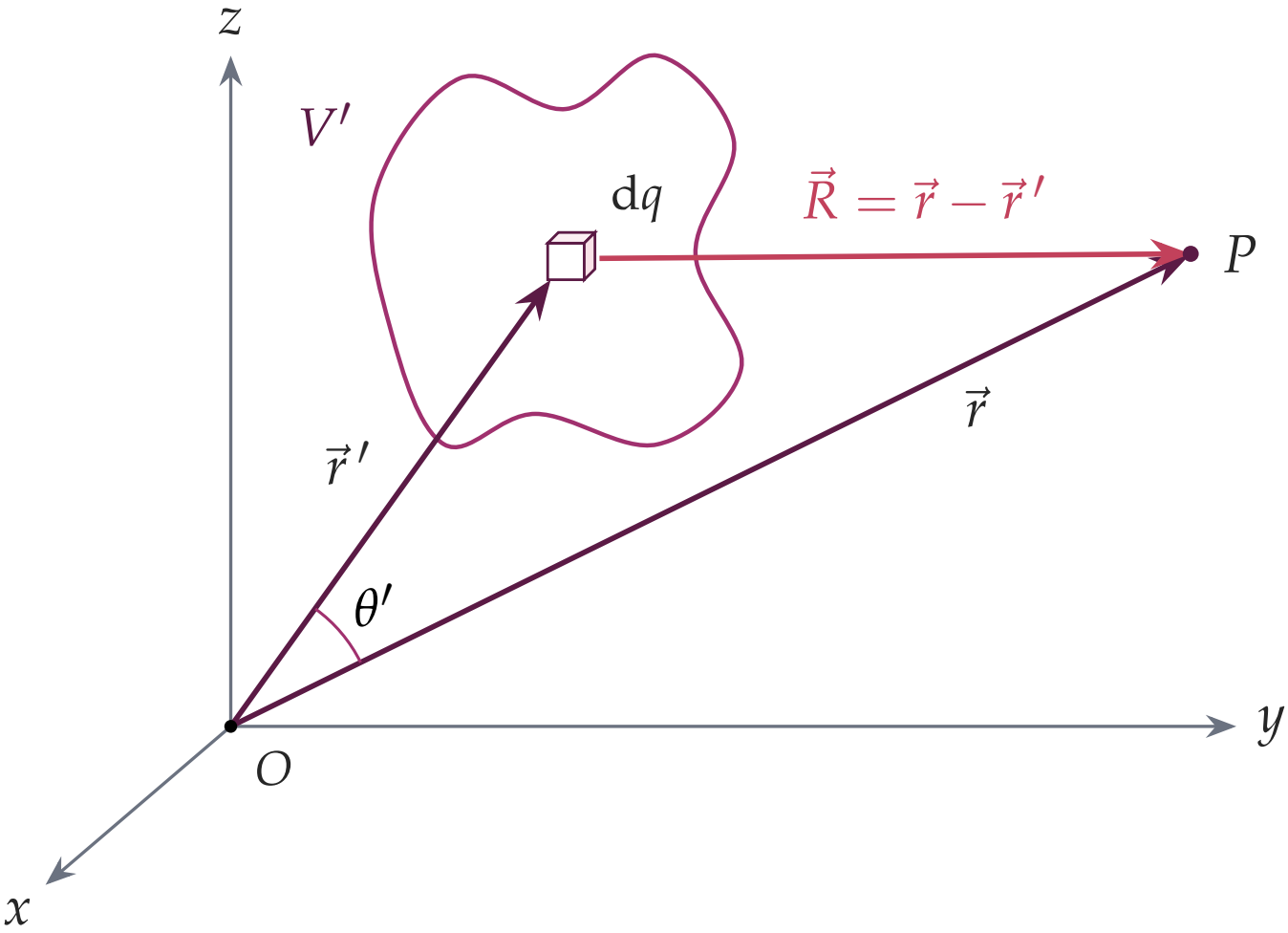

$$R^2 = r^2 + r'^2 - 2rr' \cos\theta'$$

$$\vec{r} \cdot \vec{r}\,' = rr' \cos\theta'$$